LIMIT PRESERVING TRANSFORMATIONS COMPOSE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 410-423, following the statement at 398-408.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Characterization of limit preservation. lemma-characterize-relative-limit-preserving (same chapter, card L26322, cited at ~410-411). Characterization (2) reduces the check to T = lim T_i.

Source reference. Original lemma 06BC: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 351–351. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme contained in Sch_{fppf}. Let $F, G, H : (Sch/S)^{opp}_{fppf} \to Sets$. Let $a : F \to G$, $b : G \to H$ be transformations of functors. If a and b are limit preserving, then*

$$b \circ a : F \longrightarrow H$$

is limit preserving.

Proof. Let $T = \lim_{i \in I} T_i$ as in characterization (2) of Lemma 06BC. Consider the diagram of sets

$$\begin{array}{ccc} \operatorname{colim}_i F(T_i) & \longrightarrow & F(T) \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle a} \\ \operatorname{colim}_i G(T_i) & \longrightarrow & G(T) \\ {\scriptstyle b}\downarrow & & \downarrow{\scriptstyle b} \\ \operatorname{colim}_i H(T_i) & \longrightarrow & H(T) \end{array}$$

By assumption the two squares are fibre product squares. Hence the outer rectangle is a fibre product diagram too which proves the lemma. □

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.